COMPUTER SCIENCE

LEVEL 6

DESIGN WEB APPLICATIONS

MAY/ SEP 2024 INTAKE

JANUARY 2026

JKUAT TVET INSTITTUTE – KAREN CAMPUS

PRACTICAL ASSESSMENT 1

Time: 3 HOURS

INSTRUCTIONS TO CANDIDATE:

1. You are required to perform the following tasks
   i. *Install web browser/Firefox/Microsoft edge/Opera/appropriate browser*
   ii. Install appropriate text editor/Notepad ++
   iii. *Create HTML document*
   iv. *Add basic HTML elements to file*
   v. *Use layout elements*
   vi. *Add layout attributes to file*
   vii. *Work on style sheets*
   viii. *Introduce javascript syntax*
   ix. *Work on javascript datatypes*
   x. *Work on Document Object Model(DOM) xi. Use javascript functions*
   xi. *Work on arrays*
   xii. *Install JQuery use JQuery*
   xiii. *Introduce JQuery syntax*
   xiv. *Work on JQuery events*
   xv. *Dom Manipulation with JQuery*

2. You have been provided with the following resources:
   i. *A working computer*
   ii. *Internet Access*

Task1: Installation of a web browser and text editor

You are required to perform the following activities:

i. Download web browser/Firefox/Microsoft edge/Opera/appropriate browser.
ii. Install web browser/Firefox/Microsoft edge/Opera/appropriate browser.
iii. Install the appropriate text editor.
iv. Launch the browser.
v. Launch the text editor.

Task2: ***Create HTML document and add basic elements***

i. Create HTML document that contains basic elements.
ii. Create HTML document that contain table elements
iii. Create HTML document that contain list elements
iv. HTML document that contains form elements

Task3: Create HTML document that contains layout elements shown.

i. <header>
ii. <nav>
iii. <section>
iv. <footer>

Task4: Create a HTML document to implement style sheets

i. Inline
ii. Internal
iii. External style sheet

Task5: Implement Javascript Syntax and data types

i. Demonstrate the javascript syntax
ii. Use javascript datatypes to declare variables
iii. Write a program that can add two integers and display output on the screen.

iv. Write a program that can prompt a user to input value from the keyboard and print it on the screen.

Task6: Use DOM to change elements

i. Create Javascript program that can use DOM to change elements

Task7: Use javascript functions

i. Demonstrate the javascript function syntax

ii. Use javascript datatypes to declare variables

iii. Write a program that can add two integers by the use of function.

Task8: Use Javascript arrays

i. Write statements that can demonstrate Three types of arrays.

ii. Write javascript programs that can input 5 integers into an array and print the list on the screen.

iii. Write a javascript program that can input values into 2X2 array and print the values on the screen.

Task9: Install and use JQuery(syntax and events)

i. Write the syntax of JQuery

ii. Explain the events of JQuery

iii. Write programs that implement the events of JQuery

Task10: DOM manipulation with JQuery

i. Illustrate DOM tree

ii. Write Javascript that contain objects that can be manipulated by JQuery.

Task 11: Develop a Javascript program that can be used to determine the mean grade of students in a school.

The management of a school would like to automate the processing of student marks by the implementation of a computerized sub-system. You have been tasked to develop a webpage using html and javascript that will be used to determine the mean grade of students. The sub-system should be able to capture; admission number, name, mathematics, Biology, Chemistry and physics. The system should be able to:

- Calculate the sum of all subjects,
- Calculate the average mark

48/50

COMPUTER SCIENCE

LEVEL 6

DESIGN WEB APPLICATIONS

MAY/ SEP 2024 INTAKE

JANUARY 2026

JKUAT TVET INSTITTUTE – KAREN CAMPUS

PRACTICAL ASSESSMENT 2

INSTRUCTIONS TO ASSESSOR

1. Assess the candidate as the practical progresses observing the critical areas
2. You are required to mark the practical as the candidate perform the tasks
3. You are required to take video clips at critical points
4. Ensure the candidate has a name tag and registration code at the back and front

OBSERVATION CHECKLIST

Candidate's Name & Registration Code	WAWIRA JOSHUA NYAGA TR113-C002-0003/2024		
Assessors Name & Registration Code	Lecturer Samuel Oyugi		
Venue of Assessment	TVET LAB		
Date of Assessment	10/02/2026		
Items to be Evaluated: *Please award marks as appropriate. Give a brief comment on your observation.*	**Marks Available**	**Marks Obtained**	**Comments**
TASK 1:			
1. Designed a registration form with the following information • Title • First name and Last name • Email address • Age • Password *(Award 2 mark each)*	 2 2 2 2 2	 02 02 02 02 02	 done done done done
2. Implemented gender using radio button *(Award 2 marks each)*	2	02	done
3. Implemented city using drop down listing *(Award 3 marks)*	3	03	done
4. Implemented submit button to capture the data to the database *(Award 2 marks)*	2	02	done
5. Designed a database to capture the information from the registration form • Designed a database • Assigned appropriate data type to fields *(Award 2 marks each)*	 2 2	 00 02	 undone done

6. Connected to the database *(Award 2 marks)*	2	02	done
7. Tested the functionality of the registration form. *(Award 2 marks)*	2	02	done
8. Confirmed the data is captured to the database *(Award 2 marks)*	2	02	done
9. Designed a login form with the following features			
• Form title	2	02	done
• Email address	2	02	done
• Password	2	02	done
(Award 2 marks each)			
10. Hashed the password on the login form *(Award 2 marks)*	3	03	done
11. Tested the log in page with wrong data *(Award 2 mark each)*	2	02	done
12. Designed a home page with the following information			
• a welcome message	2	02	done
• basic information	2	02	done
• a log out option	2	02	done
(Award 2 marks each)			
13. logged in to the home page successfully *(Award 3 marks)*	3	03	done
14. Logged out from the webpage successfully *(Award 3 marks)*	3	03	done
GRAND TOTAL	**50**	48	96%